Unit: 6 Unit 2: Exponents Radicals
Topic: Rationalizing Denominator

Name: ________________
Date: ____________

Homework (Jalapeno)

Q1. The ancient Egyptians used $\frac{1}{\sqrt{2}}$ to represent a fraction. Rationalize the denominator.

(A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{2}{2\sqrt{2}}$

Q2. The Mayans used $\frac{3}{\sqrt{5}}$ in their calendar calculations. Rationalize the denominator.

(A) $\frac{3\sqrt{5}}{5}$
(B) $\frac{3}{5\sqrt{5}}$
(C) $\frac{\sqrt{5}}{3}$
(D) $\frac{5}{3\sqrt{5}}$
(E) $\frac{3}{\sqrt{25}}$

Q3. The ancient Greeks rationalized $\frac{2}{\sqrt{3}}$ as $\frac{2\sqrt{3}}{3}$. What is the value of $\frac{2\sqrt{3}}{3}$ if $\sqrt{3} = 1.73$?

(A) 1.15
(B) 2.31
(C) 3.46
(D) 1.73
(E) 2.07

Q4. The Babylonians used $\frac{5}{\sqrt{6}}$ in their astronomical observations. Rationalize the denominator.

(A) $\frac{5\sqrt{6}}{6}$
(B) $\frac{5}{6\sqrt{6}}$
(C) $\frac{\sqrt{6}}{5}$
(D) $\frac{6}{5\sqrt{6}}$
(E) $\frac{5}{\sqrt{36}}$

Q5. The ancient Chinese used $\frac{4}{\sqrt{8}}$ in their mathematical texts. Rationalize the denominator.

(A) $\frac{4\sqrt{2}}{4}$
(B) $\frac{4}{4\sqrt{2}}$
(C) $\frac{\sqrt{8}}{4}$
(D) $\frac{8}{4\sqrt{8}}$
(E) $\frac{4}{\sqrt{16}}$

Q6. The ancient Indians used $\frac{6}{\sqrt{10}}$ in their mathematical discoveries. Rationalize the denominator.

(A) $\frac{6\sqrt{10}}{10}$
(B) $\frac{6}{10\sqrt{10}}$
(C) $\frac{\sqrt{10}}{6}$
(D) $\frac{10}{6\sqrt{10}}$
(E) $\frac{6}{\sqrt{100}}$

Q7. The ancient Sumerians used $\frac{8}{\sqrt{12}}$ in their economic transactions. Rationalize the denominator.

(A) $\frac{8\sqrt{3}}{6}$
(B) $\frac{8}{6\sqrt{3}}$
(C) $\frac{\sqrt{12}}{8}$
(D) $\frac{12}{8\sqrt{12}}$
(E) $\frac{8}{\sqrt{144}}$

Q8. The ancient Phoenicians used $\frac{9}{\sqrt{15}}$ in their maritime trade. Rationalize the denominator.

(A) $\frac{9\sqrt{15}}{15}$
(B) $\frac{9}{15\sqrt{15}}$
(C) $\frac{\sqrt{15}}{9}$
(D) $\frac{15}{9\sqrt{15}}$
(E) $\frac{9}{\sqrt{225}}$

Open Ended Questions (FRQ)

Q9. The ancient Egyptians built the Great Pyramid using $\frac{10}{\sqrt{20}}$ as a ratio of base to height. Rationalize the denominator and simplify.

Q10. The ancient Aztecs used $\frac{12}{\sqrt{24}}$ in their architectural designs. Rationalize the denominator and simplify.

Chef's Tips

- Tip 1: Ensure students understand the historical context of rationalizing denominators.
- Tip 2: Encourage students to simplify their answers.
- Tip 3: Remind students to check their work for accuracy.